

GRADE 7 · READING INFORMATIONAL TEXT · RI.7.2

Central Ideas and Summary

Determine two or more central ideas in an informational text about urban heat islands, analyze how each develops across the text, and write an objective summary.

Lesson Map · Passage · Guided · Partner · Independent Practice · Small Group · Answer Key

What You Will Learn Today

Learning Goals

Understand RI.7.2: Determine two or more central ideas in an informational text about urb...

Read an original passage closely

Support every answer with text evidence

Work through guided, partner, and independent practice

How the Lesson Runs

Warm-up and passage

Guided practice together

Partner discussion

Independent + small-group work

Exit ticket and answer key

Unpacking the Standard

A central idea is a main point the text develops with evidence, not just a topic. Informational texts often develop two or more central ideas at once. An objective summary reports only what the text says, in order, without opinions or added information.

Modeled Example

Topic: cities are hot. Possible central idea?

Urban surfaces like asphalt and concrete absorb and re-release heat, making cities significantly warmer than surrounding rural areas — a specific, evidence-supported claim, not just a topic label.

Is "the study measured temperatures in twelve neighborhoods" an objective summary sentence or an opinion?

Objective — it reports a factual detail from the text without added judgment.

The City That Doesn't Cool Down

1. On a July afternoon, a thermometer in downtown Meridian reads 97 degrees. Twelve miles away, in the farmland outside the city, a matching thermometer reads 89 degrees. The two locations sit under the same sun, at the same time, on the same day.
2. This gap has a name: the urban heat island effect. Cities tend to run several degrees warmer than their surrounding rural areas, especially in the late afternoon and evening, because of how urban surfaces interact with heat.
3. Asphalt and dark rooftops absorb far more solar energy than grass or soil does. Through the day, these surfaces heat up; after sunset, they slowly release that stored heat back into the air, keeping nighttime temperatures elevated long after rural areas have cooled.
4. A second factor compounds the first: cities have far less vegetation. Trees and plants cool their surroundings through a process called evapotranspiration, releasing water vapor that absorbs heat from the air. A parking lot does the opposite, offering no cooling effect at all.
5. Meridian's planning department measured temperatures across twelve neighborhoods over one summer. The three neighborhoods with the least tree cover averaged 6.2 degrees warmer than the three neighborhoods with the most tree cover, even though all twelve sites were within the same city.

The City That Doesn't Cool Down, continued

6. The effect is not evenly distributed. Older, tree-lined neighborhoods with large yards showed temperatures close to the surrounding countryside. Newer developments, built with more pavement and fewer mature trees, ran consistently hotter.

7. City planners have started requiring lighter-colored roofing on new buildings and setting minimum tree-planting requirements for new developments. Early results from one pilot block, repaved with a reflective coating, show afternoon surface temperatures roughly 10 degrees cooler than the untreated block next to it.

Warm-up · Before You Read

1. What two locations does paragraph 1 compare, and what is the temperature difference?
2. State one central idea developed in this text.

Guided Practice

3. State a second, different central idea developed in this text.

4. How does paragraph 3 develop the first central idea (about urban surfaces)?

Guided Practice

5. How does paragraph 4 introduce the second central idea (about vegetation)?

6. What specific evidence in paragraph 5 supports the vegetation-related central idea?

Partner Discussion

7. How does paragraph 6 further develop the central idea about vegetation?

8. How does paragraph 7 connect back to both central ideas?

Partner Discussion

9. What specific result does the pilot block in paragraph 7 provide as evidence?

Independent Practice

10. Write one objective sentence summarizing paragraphs 1–2 (no opinions).

11. Write one objective sentence summarizing paragraph 5 (no opinions).

Independent Practice

12. Identify one word in this draft summary that is an opinion, not a fact: "The city's shockingly poor planning caused dangerous heat levels downtown."

13. Write an objective one-sentence summary of paragraph 7.

Independent Practice

14. Why is "the city clearly does not care about its hottest neighborhoods" not appropriate for an objective summary?

15. Combine paragraphs 2, 3, and 4 into one two-sentence objective summary.

Small-Group Work

16. How do the two central ideas (surface absorption and vegetation) work together to explain the heat island effect?

17. Trace how the vegetation-related central idea develops from paragraph 4 to paragraph 6.

Small-Group Work

18. Is "reflective coating lowered surface temperature by 10 degrees on the pilot block" objective or subjective? Explain.

19. Write a two-sentence objective summary of the whole text.

Exit Ticket

20. State both central ideas in one sentence each, and support each with one piece of cited evidence.

Answer Key (1 of 4)

1. Downtown Meridian at 97 degrees and farmland twelve miles away at 89 degrees, an 8-degree difference.
2. Urban surfaces like asphalt and dark rooftops absorb and slowly release heat, making cities warmer than surrounding rural areas.
3. The amount of vegetation in a neighborhood significantly affects how much the urban heat island effect impacts that specific area.
4. It explains the mechanism: asphalt and dark rooftops absorb solar energy during the day and slowly release it at night, keeping temperatures elevated even after sunset.
5. It explains that trees and plants cool their surroundings through evapotranspiration, while paved surfaces like parking lots provide no such cooling effect.

Answer Key (2 of 4)

6. Meridian's planning department found the three neighborhoods with the least tree cover averaged 6.2 degrees warmer than the three with the most tree cover.

7. It shows the effect is uneven within the same city — older, tree-lined neighborhoods stay close to rural temperatures, while newer, more paved developments run consistently hotter.

8. It describes solutions addressing both: lighter-colored roofing addresses the surface-heat-absorption idea, and tree-planting requirements address the vegetation idea.

9. A block repaved with reflective coating showed afternoon surface temperatures roughly 10 degrees cooler than the untreated block next to it.

10. A downtown location can be several degrees warmer than nearby rural areas on the same day, an effect known as the urban heat island effect.

Answer Key (3 of 4)

11. A study of twelve Meridian neighborhoods found that areas with the least tree cover averaged 6.2 degrees warmer than areas with the most tree cover.

12. Shockingly (or dangerous) — these are judgments about the situation, not observable facts stated in the text.

13. City planners have begun requiring lighter roofing and tree-planting in new developments, and a pilot block with reflective coating showed a 10-degree temperature drop.

14. It is an opinion and an assumption about intent that is not directly stated or supported by the text, which only reports temperature data and city actions.

15. The urban heat island effect makes cities warmer than surrounding rural areas because surfaces like asphalt and dark rooftops absorb and slowly release heat. Cities also have less vegetation, which reduces the natural cooling effect trees and plants provide.

Answer Key (4 of 4)

16. Both factors combine to raise urban temperatures: heat-absorbing surfaces add warmth while reduced vegetation removes a natural cooling process, making their combined effect stronger than either alone.

17. Paragraph 4 introduces the general mechanism (trees cool through evapotranspiration); paragraph 6 provides specific evidence that neighborhoods with more trees stay cooler, showing the idea moving from explanation to real-world data.

18. Objective — it is a specific, verifiable result stated in the text, without added judgment.

19. Urban areas can run several degrees warmer than nearby rural areas because heat-absorbing surfaces and reduced vegetation combine to raise temperatures, an effect studied in Meridian across twelve neighborhoods. City planners are now testing solutions like reflective roofing and tree-planting requirements, with early results showing notable temperature reductions.

20. Answers vary; require two distinct, correctly stated central ideas (not just topics) plus one specific, correctly cited detail supporting each.

Exit Ticket

Answer on a sticky note

Answer the exit-ticket question from today's lesson.

Cite at least one piece of evidence from the passage.

Remember

Name your claim or answer

Quote or cite the evidence

Explain how it supports your answer